HYPOXIA WORKSTATION





Model: H-300

BiOxia

www.bioxia.co.in

Hypoxia Workstation H-300

Body Construction:

- Manufactured from durable, non-corrosive acrylic resin sheet, providing excellent visibility, chemical resistance, and thermal insulation.
- Designed for accurate and stable control of oxygen (O_2), carbon dioxide (CO_2), temperature, and humidity.

Dimensions (L × H × D) – Approx.

- External Dimensions: 950 × 740 × 720 mm
- Main Chamber: 600 × 570 × 540 mm
- Interlock / Airlock: 180 × 200 × 200 mm

Main Workstation Capacity:

- Approx. 300-litre incubation volume
- Up to 350 plates of 90mm X 15mm
- Additional space for routine inoculation and handling operations
- One internal power supply socket inside the chamber

Interlock / Airlock Capacity:

- Integral 12-litre airlock enabling rapid transfer of cell ware within 20 seconds
- Up to 25 plates of 90mm X 15mm
- Automatic interlock / Airlock cycle for efficient operation

Safety Features:

- Excess pressure relief valve
- Temperature overshoot protection controller
- Inbuilt condensate control technology
- Automatic forced purging system for main chamber oxygen control with adjustable purging period
- Additional internal air-direction system for faster oxygen reduction
- Fastest oxygen control capability: rapid reduction from atmospheric O_2 to 0.1%
- Auto excess pressure control system

Display & Control System:

- LCD Touch Screen Interface with 4-level password protection
- Continuous display and monitoring of:
    - Chamber pressure
    - Temperature
    - Oxygen percentage
    - Internal socket status
    - Positive chamber pressure (0.15 kPa / 0.0015 bar)
    - Fluorescent light status
- Visual alarms provided for:
    - Low or excess oxygen levels
    - Gas cylinder pressure
    - Chamber pressure and temperature deviations
- Alarms and status notices clearly displayed on the touch screen
- Remote access and control via touch screen interface

Said Touch Screen display continuously indicate & controls:

- Controls O2 from 0.1% to 20.0%in 0.1% increments
- Controls CO2 from 0.1% to 30.0%in 0.1% increments
- Sensors provided for real-time monitoring of O_2 and CO_2 concentrations

Said Touch screen simplifies the calibration process & Data Management:

- One-touch automatic O_2 sensor calibration
- Rapid atmospheric equalization
- Data download facility
- Internal memory storage for minimum 31 days
- Faculty recognition feature

Said Touch Screen display allows hypoxic cycling:

- Programmable cycling for up to 4 different oxygen concentrations
- User-defined time sequence operation
- The sensors remain inside the workstation for real time monitoring of O2 & Co2 concentration

Pressure & Ergonomic Design:

- Continuous positive pressure maintenance (0.15 kPa / 0.0015 bar)
- Front viewing panel with 10% slope to enhance visibility and reduce glare
- Circular or Oval glove ports suitable for bare-hand or easy gloved operation, ensuring maximum operator comfort

Temperature & Humidity Control:

- Heating element along with controller shall be provided to maintain the temperature inside chamber within 5°C above ambient to 45°C
- Temperature uniformity: ±1°C at 37°C
- Inbuilt trim-point for temperature calibration
- Relative humidity controls up to 80% RH with display
- Heater provided for dehumidification

Utilities & Accessories:

- Utility port with flexible hose for manual nitrogen purging inside the chamber
- Separate purging valves for main chamber and interlock chamber
- Activated detox system for odor absorption
- Inspection spot lamp

Gas & Operation:

- Inlet provision for two separate gas cylinders (N_2 & O_2)
- Fully integrated gas mixing system (external gas mixer not required)
- System operation through wireless radio-controlled foot switch for:
    - Sleeve vacuum control
    - Gas purging
    - Spot lamp operation

Supplied Accessories:

- Pair of gloves
- Internal full-length shelf
- O_2 Sensor & CO_2 Sensor
- Sleeve kit
- Glove retaining O-ring
- Evaporator
- Airlock transfer tray
- Wrist seals
- Dry-lube lubricating powder
- Fuse set
- User manual
- Assorted components for mains and gas connections

Gas Supply:

- System supplied with filled N_2, CO_2, and O_2 cylinders along with regulators

Power Supply:

- Input Voltage: 220–240 V AC, 50 Hz
- Supplied with power plug
- Resettable over-current protection breaker fitted

Certifications:

- ISO 9001 : 2015 Certified
- CE Marked : CE marked product
- GMP Certified
- ISO 14001:2015
- ISO 27001:2022

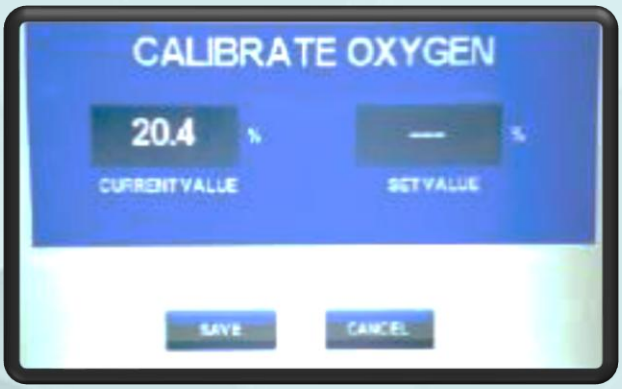
Touch Screen Display

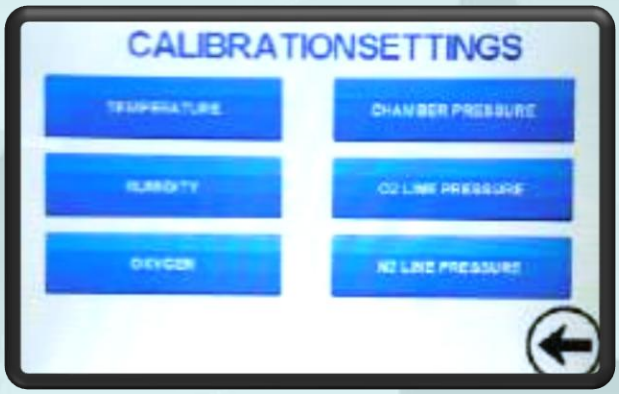


Data Logging Facility

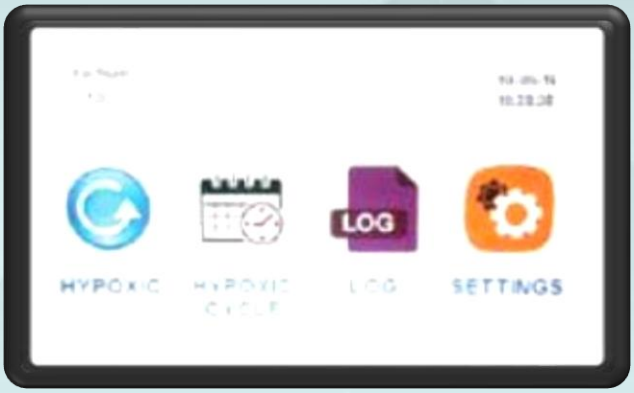


Calibration of Oxygen





Settings



"Flexible hose" One Utility Connection Port



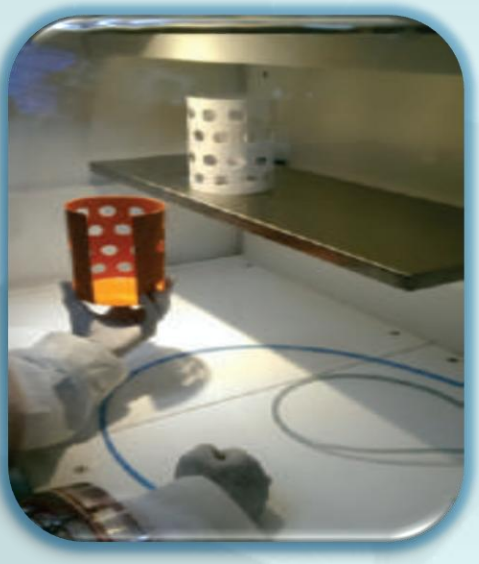
Interlock Transfer



Condensing Cooling Plate



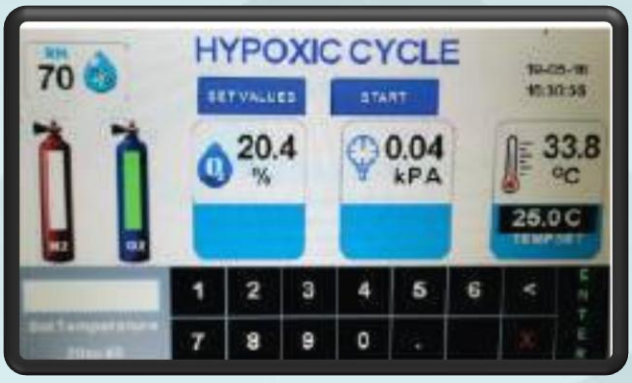
Spot Lamp



Foot Switch



Hypoxic Cycle



Petri Dish Holders



PRODUCT SUMMARY

MODEL	H-300
Interlock Cycle Time	20s
Cabinet Mode of Construction	Solvent bonded Acrylic
Max Incubation Temperature	45°C-50°C
Gas Supply	Inlet operation for four separate gases Premixed anaerobic gas (H2: Co2: N2) (10% : 10 % : 80 %), N2, Co2 and Air or Oxygen.
Humidity control	√
Condensate Control	√
Positive Pressure Display	√
Safety Excess Pressure Relief Valve	√
Low Gas Alarm	√
Inspection Spot Lamp	√
Fluorescent Illumination	√
Palladium Catalyst	√
Activated Carbon Particles	√
Oxygen Sensor with Display	√
Oxygen Controlled with Display 0.1% to 20.0%	√
Co2 Controlled with Display 0.1% to 30.0%	√
Ultra Violet (UV) Light	O
Bare Hand Facility	√
Incubator	O
Foot Switch Operation	√
Removable Front	O
Internal Power Supply	√
Utility nozzle Connection	√
LCD Display with control	√
Fan Unit	√
Adjustable Shelf	√
Remote Control Facility	√
Cabinet Stand	O
Regulator for Cylinder	√
Gloves	√
Vacuum Line Port	√
Cable Gland Port	√
Petri Dish Holder	√

√ - Available O - Optional N/A - Not Available

BIOXIA

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